MARLIN

Marine Adaptive Response and Lifecycle Intervention Network



A post-alert response for seasonal jellyfish blooms

A testable, two-layer concept focused on marine-life protection and less manual response.

TEAM Bashger Aljneibi · Keyal Doshi · Abdul Fattah Mohammed · Abdul Bari Mohammed

Two physical layers, one simple objective

Respond after the existing bloom alert—guide, collect gently, and keep the intake screens as backup.

97.5%

delivered to a hypothetical supervised receiver

2.5% remained uncaptured in the v4 reference run.

SCREENING RESULT

A model outcome under stated assumptions—not a field catch-rate claim.

01 **Outer guide**

Primary interception; guided conveyance to a wet receiver.

02 **Second guide**

Independent reserve line for bloom that passes the first.

03 **Measured response**

Operate after the ENEC alert; local interlocks bound automation.

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Reference run: 12 m assumed depth · 200 m corridor · 1,200 kg/h synthetic arrival · 8 h simulation.

The hard part begins after the alert

MARLIN is a bloom-response concept, not a replacement for ENEC's offshore warning program.

A large bloom is a moving, high-volume load.

Manual clearing can require sustained field effort. Any new barrier or collector must also avoid shifting risk onto other marine life or water flow.

01

Reduce repeated manual work

Automate routine conveyance and monitoring, with operators retaining oversight.

02

Protect marine life

Measure bycatch, injury, escape routes, and handling outcomes before scaling.

03

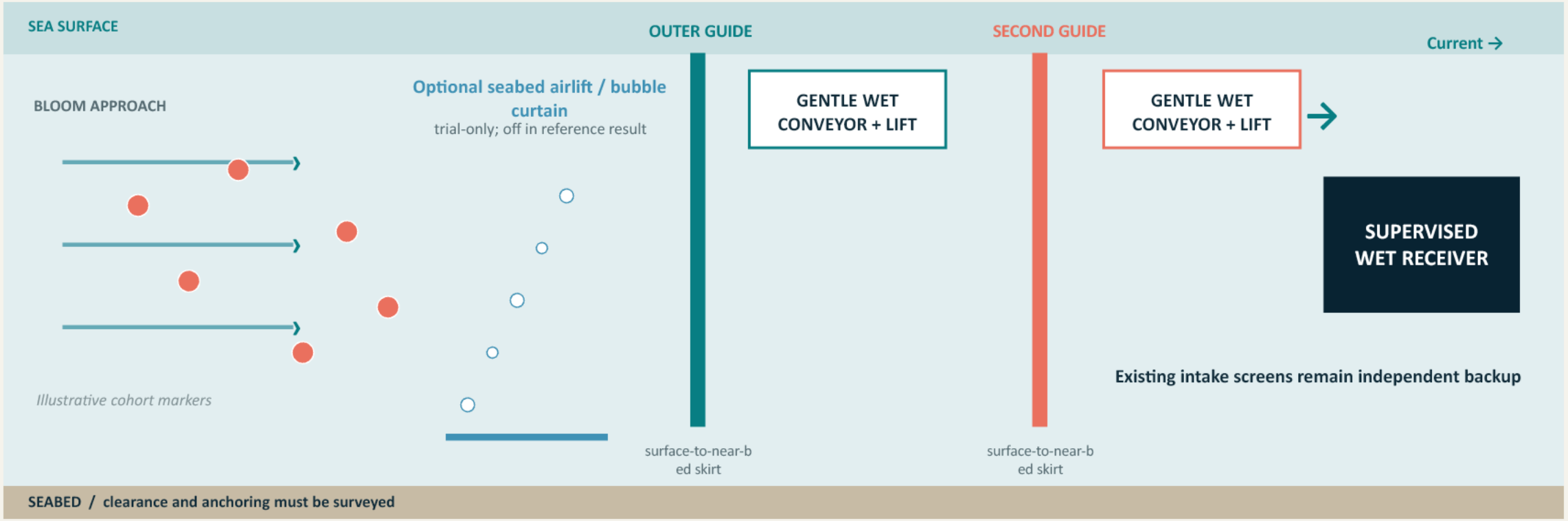
Handle continuous volume

Size collection, buffers, and response capacity for bloom loads—not isolated animals.

Challenge brief: post-alert management • marine-life protection • lower manual intervention • cost-conscious scaling.

Two interception layers; one controlled collection route

Conceptual cross-section only. Final placement, coverage, skirt depth, and collection points require site data.



The model assumes 12 m water depth; actual bathymetry and side/bottom escape paths are unknown.

Automation inside hard limits

Start on an existing alert. Use local sensing to adjust the response, with operator oversight and independent fail-safe states.

01 / ALERT

ENEC bloom warning

Response window opens

02 / SENSE

Flux + flow + load

Local instruments confirm conditions

03 / ACT

Outer → second guide

Convey and lift within verified limits

04 / VERIFY

Receiver + escape count

Report delivered, held, and missed mass

CONTROL STACK

AI / ML (future, advisory)

Predict bloom flux and recommend activation or setpoints after site data exist. No trained ML/CV model is in the current simulation.

Local PLC + interlocks

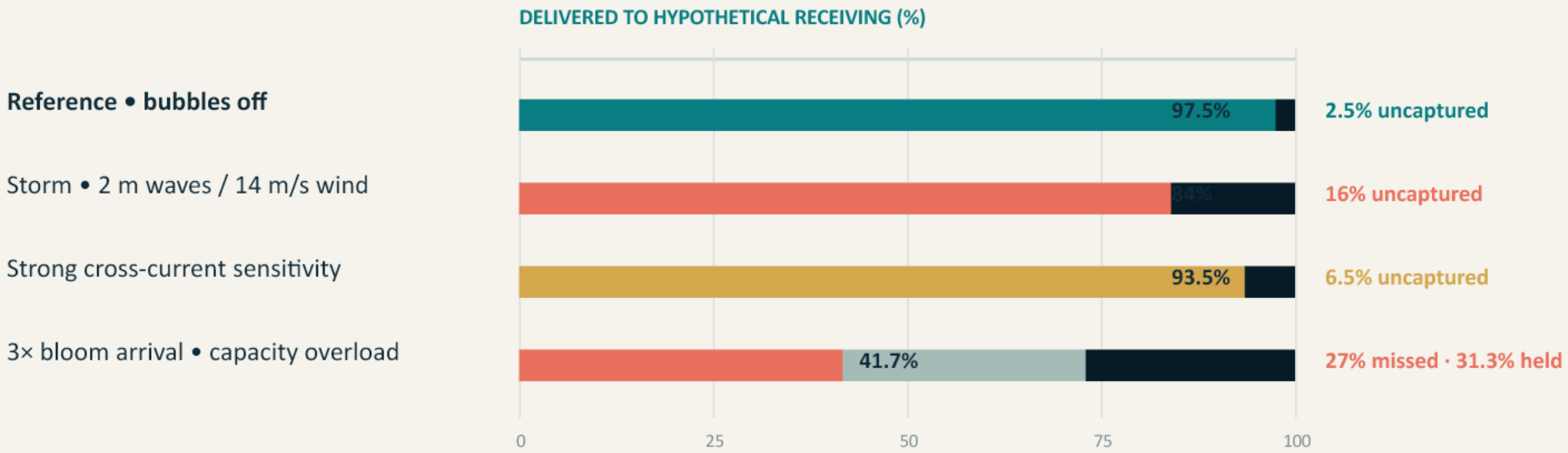
Enforce depth, speed, pressure, load, and stop limits. Local operation remains available if communications fail.

Operator authorization

Supervised first deployment. Animal identification, handling, and release decisions follow an approved protocol.

Strong reference case; clear stress-case misses

Bars show share delivered to the model's hypothetical receiver—not biological survival or plant performance.



WHAT THIS SUPPORTS

The architecture is worth physical testing. The 95% delivery goal appears only in the synthetic reference case; wind, overload, cross-flow, common-cause misses, and depth change can break it.

27 numerical/invariant checks passed; zero mass-ledger error. These verify code consistency, not field physics.

Protect the animals—and the flow around them

Treat ecological impact and hydraulic resistance as pass/fail gates, not as benefits assumed from a drawing.

01

Limit contact

Prefer open, smooth guidance surfaces and slow wet conveyance. Mesh, skirt, mooring, and shear choices need tank evidence.

02

Separate bycatch

Detect non-target organisms before handling; pause, isolate, and follow an ecologist-approved release protocol.

03

Protect water flow

Track differential pressure, flow change, line load, and clogging. Retract or bypass only within approved limits.

04

Close the ledger

Record delivered, held, escaped, bycatch, and unknown mass. No release direction is assumed safe without local flow checks.

STOP RULE

Unknown bycatch • rising load/pressure • failed sensor • unsafe sea state → isolate the section and alert the operator.

Spend energy only where it buys a measured benefit

The cost case is a design strategy now; site-specific capital and operating costs still need supplier quotes.

PASSIVE FIRST

Guide bands do the continuous interception; powered collection operates when a bloom response is active.

MODULAR

Segmented guides and independent collectors make inspection and repair local rather than system-wide.

NO ROBOT TAX IN THE MVP

Use an operator-supervised service craft during trials; defer an uncrewed vessel until it proves a labor or safety benefit.

BUBBLES ARE AN EVIDENCE-GATED OPTION

Not the core barrier

The current comparison does not show a dependable capture gain from continuous bubble operation.

77.1 **kWh / bloom event**

model estimate for two continuously operated diffuser lines; not a field energy measurement.

Test airlift and curtain independently in a flume; retain only if capture, ecology, and energy improve together.

Prove the hard parts before a site pilot

Every gate should measure capture, marine impact, flow cost, and failure recovery together.

0 CALIBRATE

Bathymetry, ADCP current, intake-adjacent flow, bloom trajectories, and operator constraints.

ADVANCE WHEN

Site envelope signed off

1 FLUME

Guide geometry, airlift/bubble on-off, conveyor speed, clogging, escape, and bycatch handling.

ADVANCE WHEN

Target <5% escape in defined test envelope

2 SHELTERED SEA

Permitted deployment with ecological observation, load cells, pressure, flow, and recovery drills.

ADVANCE WHEN

Impact and hydraulic limits met

3 CONTROLLED DEMO

Operator-led demonstration under approved boundaries; no modification to plant or nuclear-safety systems.

ADVANCE WHEN

ENEC authorizes next step

Core measures Escape % · bycatch / injury · pressure & flow change · kWh per tonne · staff-hours · recovery time · mass-balance closure

A promising screen—not a promise

The model helps decide what to test next. It cannot yet certify a safe, 95% field outcome.

KEY ASSUMPTIONS

- Synthetic bloom arrivals at 1,200 kg/h for four hours; not a public local abundance record.
- 12 m flat water depth, 200 m corridor, and guide contact probabilities of 0.90 / 0.80 are uncalibrated.
- Weather and circulation use public regional reanalysis; forcing does not resolve local intake flow, turbulence, or guide feedback.
- No validated ML/CV, bycatch/injury model, animal survival model, or field-tested handling/release protocol.

NEXT DECISION

Request a bounded technical review

Confirm deployment boundary, depth/current envelope, allowable flow change, marine-life review pathway, and approved response handling. Then calibrate the flume model and retest its worst cases.

Declaration: AI tools supported drafting and visual layout; the team reviewed the assumptions and owns the proposal.

SOURCES

ATP 2026 ENEC brief • Haberlin et al., Aquaculture (2021), DOI 10.1016/j.aquaculture.2020.735915 • Copernicus ERA5 / Marine ocean reanalysis

Public data are regional forcing inputs—not a jellyfish trajectory/abundance dataset and not site measurements.